



Department of Computer Science and Engineering  
Premier University

## CSE 459: Pattern Recognition

Title: Regression Analysis and Model Comparison for California  
Housing Price Prediction

**Submitted by:**

|                        |                               |
|------------------------|-------------------------------|
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**Remarks**

# Regression Analysis and Model Comparison for California Housing Price Prediction

## 1 Dataset Selection

The California Housing Dataset was selected for this regression task. It is available through the Scikit-learn library and represents a real-world problem of predicting housing prices.

### Dataset Overview

| Attribute  | Description                                                                    |
|------------|--------------------------------------------------------------------------------|
| Source     | Scikit-learn dataset repository                                                |
| Features   | Income, house age, rooms, bedrooms, population, occupancy, latitude, longitude |
| Target     | Median house value                                                             |
| Total Rows | 20,640                                                                         |

### 1.1 Sample Data Instances

| Index | MedInc | HouseAge | AveRooms | AveBedrms | Population | AveOccup | Lat   | Long    | Target |
|-------|--------|----------|----------|-----------|------------|----------|-------|---------|--------|
| 0     | 8.3252 | 41       | 6.98     | 1.02      | 322        | 2.56     | 37.88 | -122.23 | 4.526  |
| 1     | 8.3014 | 21       | 6.24     | 0.97      | 2401       | 2.11     | 37.86 | -122.22 | 3.585  |
| 2     | 7.2574 | 52       | 8.29     | 1.07      | 496        | 2.80     | 37.85 | -122.24 | 3.521  |
| 3     | 5.6431 | 52       | 5.82     | 1.07      | 558        | 2.55     | 37.85 | -122.25 | 3.413  |
| 4     | 3.8462 | 52       | 6.28     | 1.08      | 565        | 2.18     | 37.85 | -122.25 | 3.422  |

**Explanation:** The table presents sample records from the dataset, showing how input features and the target variable are structured. It helps in understanding data variation and feature ranges.

## 2 Dataset Visualization

### 2.1 Feature Distribution

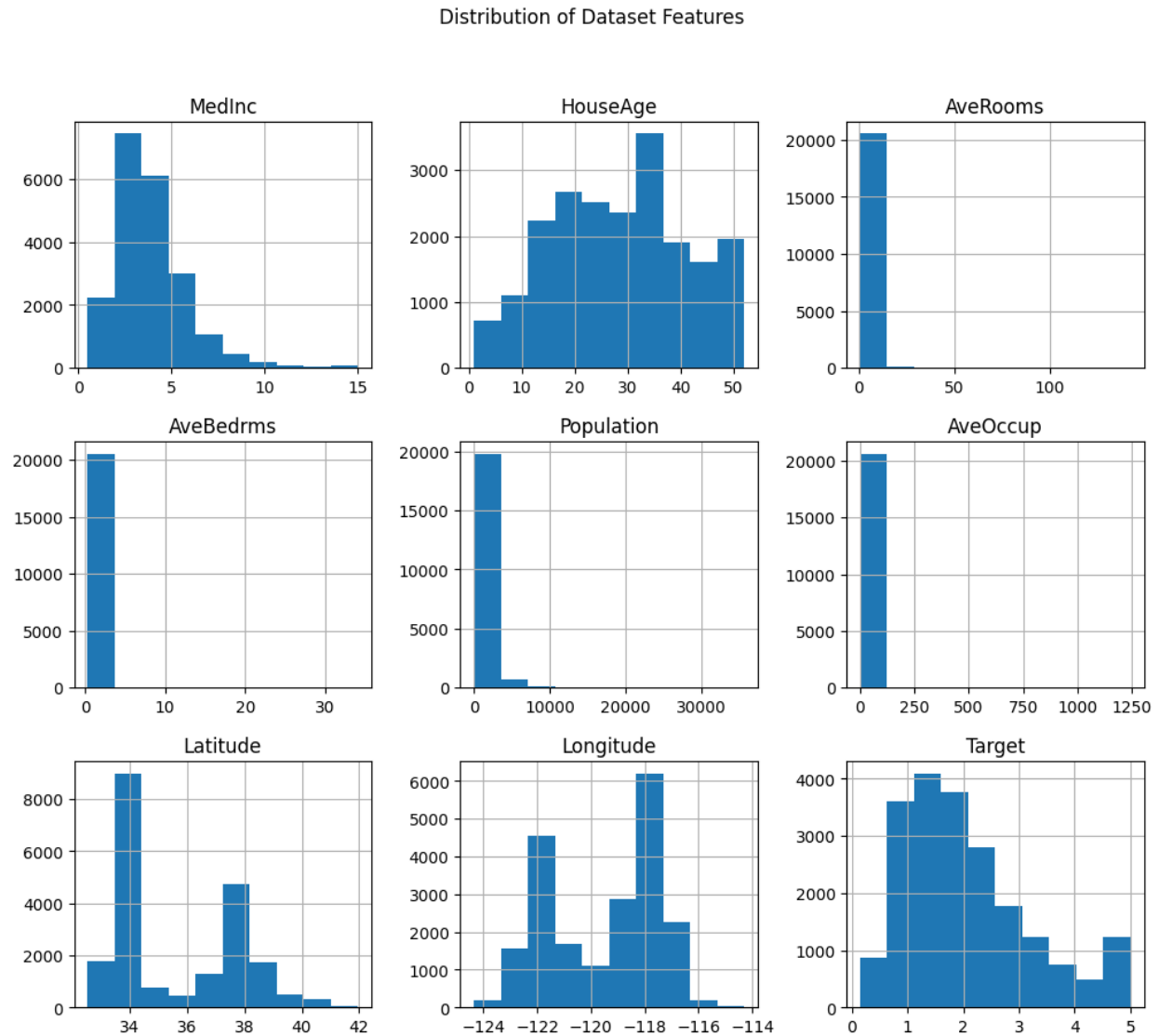


Figure 1: Distribution of Dataset Features

**Explanation:** The histogram shows the distribution of all features, highlighting differences in scale and skewness. This justifies preprocessing steps such as feature scaling.

## 3 Data Preprocessing

### 3.1 Missing Values

No missing values were found.

## 3.2 Feature Scaling

StandardScaler was applied.

**Reason:** Scaling ensures all features contribute equally and improves Linear Regression performance.

## 3.3 Train-Test Split

80% training and 20% testing split ensures fair evaluation.

# 4 Model Implementation

## 4.1 Code

```
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor

data = fetch_california_housing()
X, y = data.data, data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

lr = LinearRegression()
rf = RandomForestRegressor(n_estimators=100, random_state=42)

lr.fit(X_train_scaled, y_train)
rf.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test_scaled)
y_pred_rf = rf.predict(X_test)
```

**Explanation:** The dataset is loaded and split into training and testing sets. Feature scaling is applied for Linear Regression. Two models are trained: Linear Regression (linear model) and Random Forest (ensemble model). Predictions are generated using both models.

## 5 Model Evaluation

```
from sklearn.metrics import mean_absolute_error, mean_squared_error,
    r2_score
import numpy as np

def evaluate(y_test, y_pred):
    mae = mean_absolute_error(y_test, y_pred)
    mse = mean_squared_error(y_test, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_test, y_pred)
    return mae, mse, rmse, r2
```

**Explanation:** This function computes evaluation metrics including MAE, MSE, RMSE, and  $R^2$ , which measure prediction error and model performance.

### 5.1 Results

| Model             | MAE  | MSE  | RMSE | $R^2$ |
|-------------------|------|------|------|-------|
| Linear Regression | 0.53 | 0.55 | 0.74 | 0.58  |
| Random Forest     | 0.33 | 0.25 | 0.50 | 0.81  |

**Analysis:** Random Forest shows lower errors and higher  $R^2$ , indicating better predictive performance.

## 6 Visualization and Analysis

### 6.1 Actual vs Predicted Values

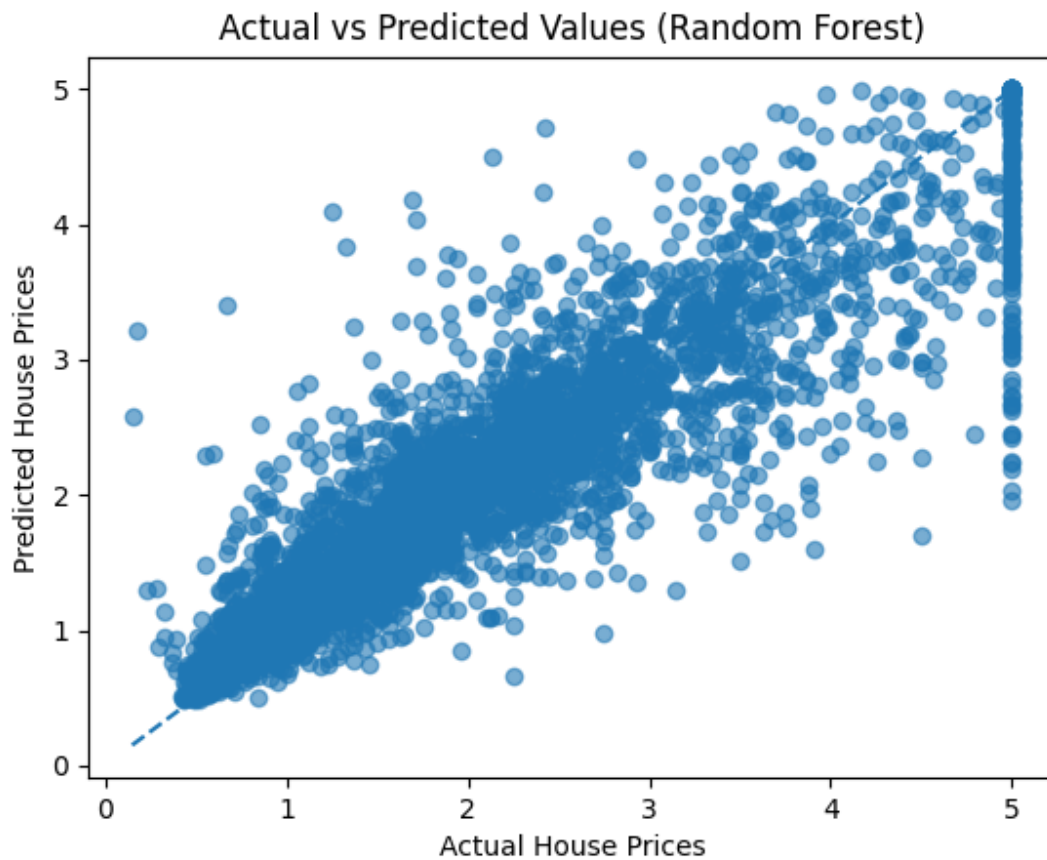


Figure 2: Actual vs Predicted House Prices

**Explanation:** Points close to the diagonal indicate accurate predictions. The model captures the overall trend effectively.

## 6.2 Actual vs Predicted (Line Comparison)

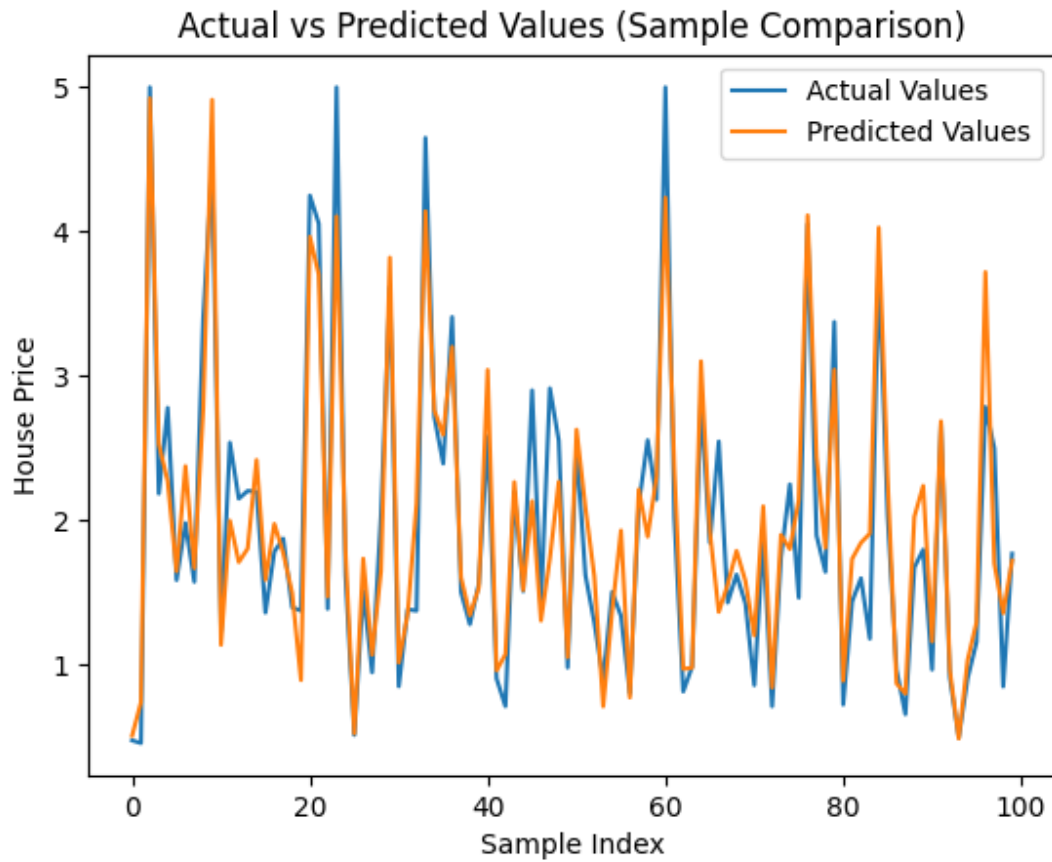


Figure 3: Actual vs Predicted Comparison

**Explanation:** Predicted values closely follow actual values, showing consistent performance.

## 6.3 Feature Importance

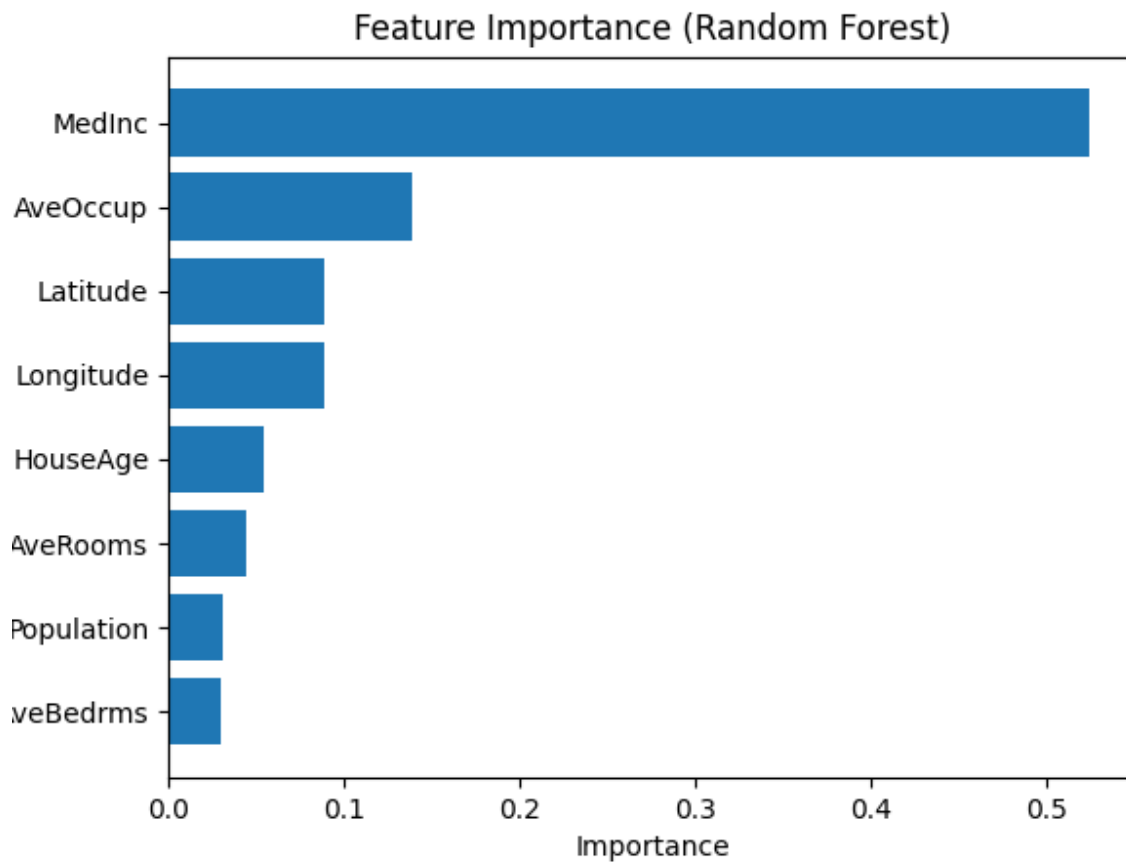


Figure 4: Feature Importance

**Explanation:** Median income has the highest importance, indicating strong influence on house prices.

## 7 Final Model Justification

Random Forest Regression is selected as the best-performing model.

- Captures non-linear relationships
- Handles feature interactions
- Provides higher accuracy

## 8 Conclusion

The report demonstrates a complete regression analysis workflow. Random Forest outperforms Linear Regression due to its ability to model complex patterns and provide more accurate predictions.